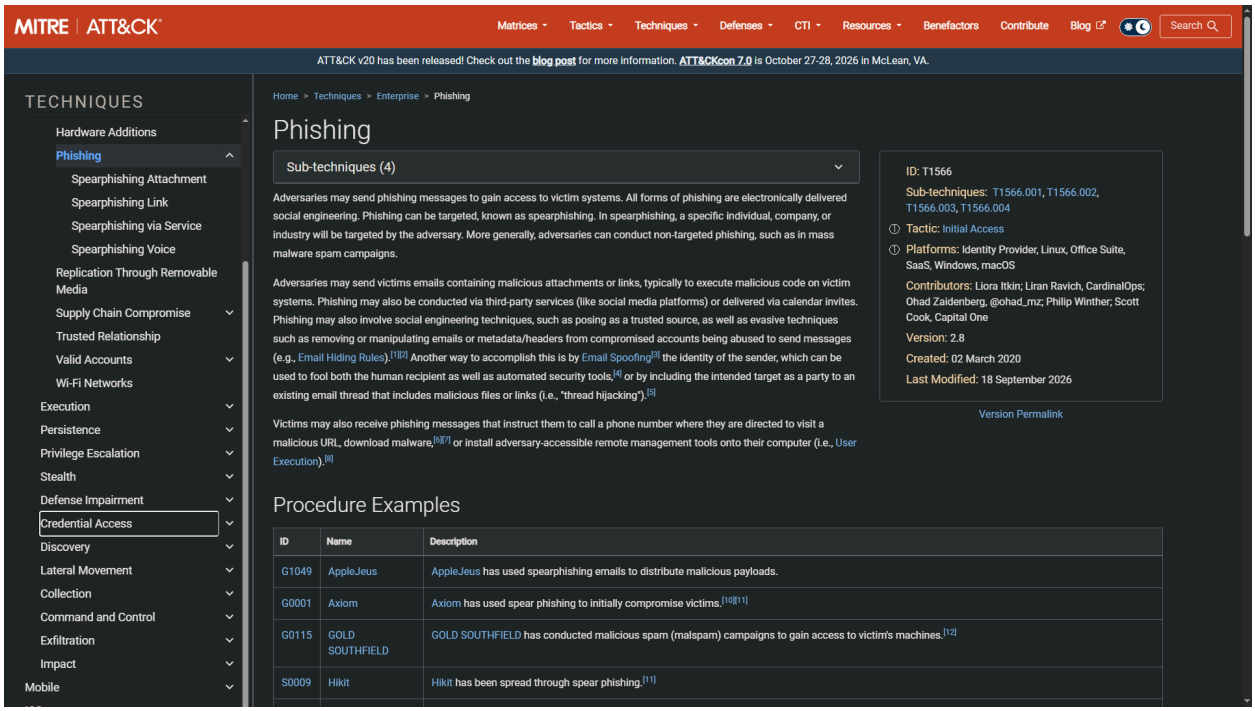
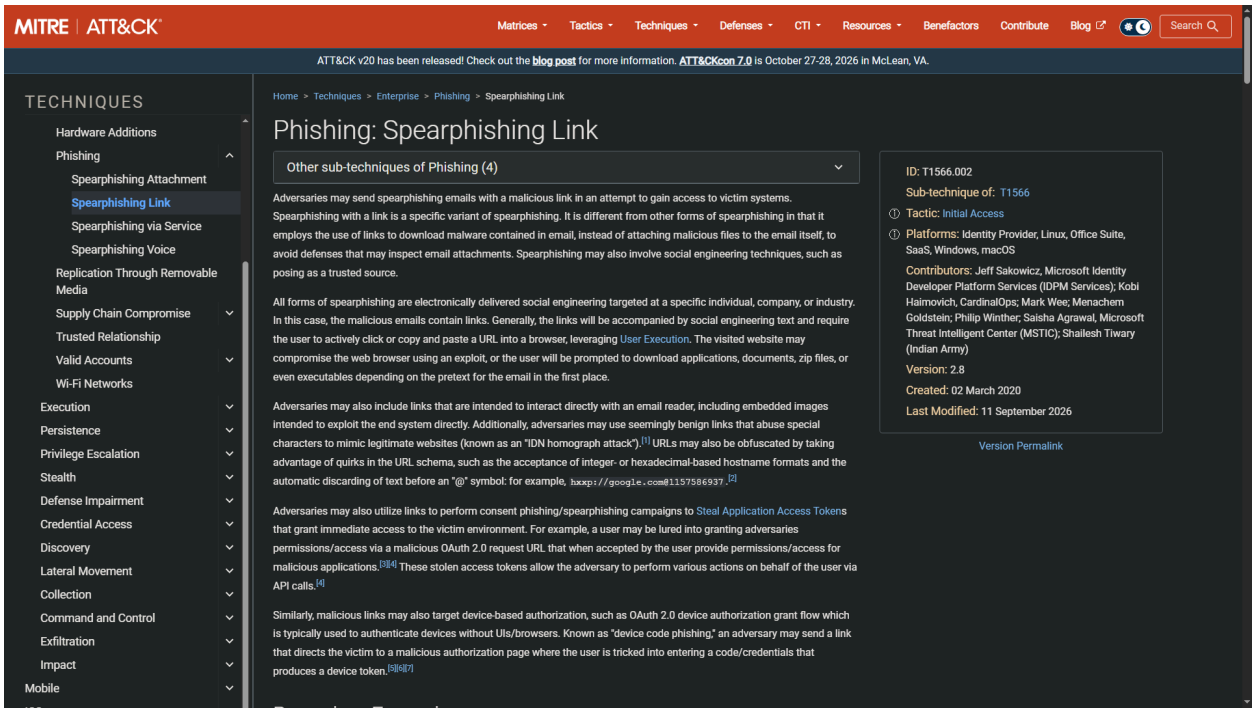
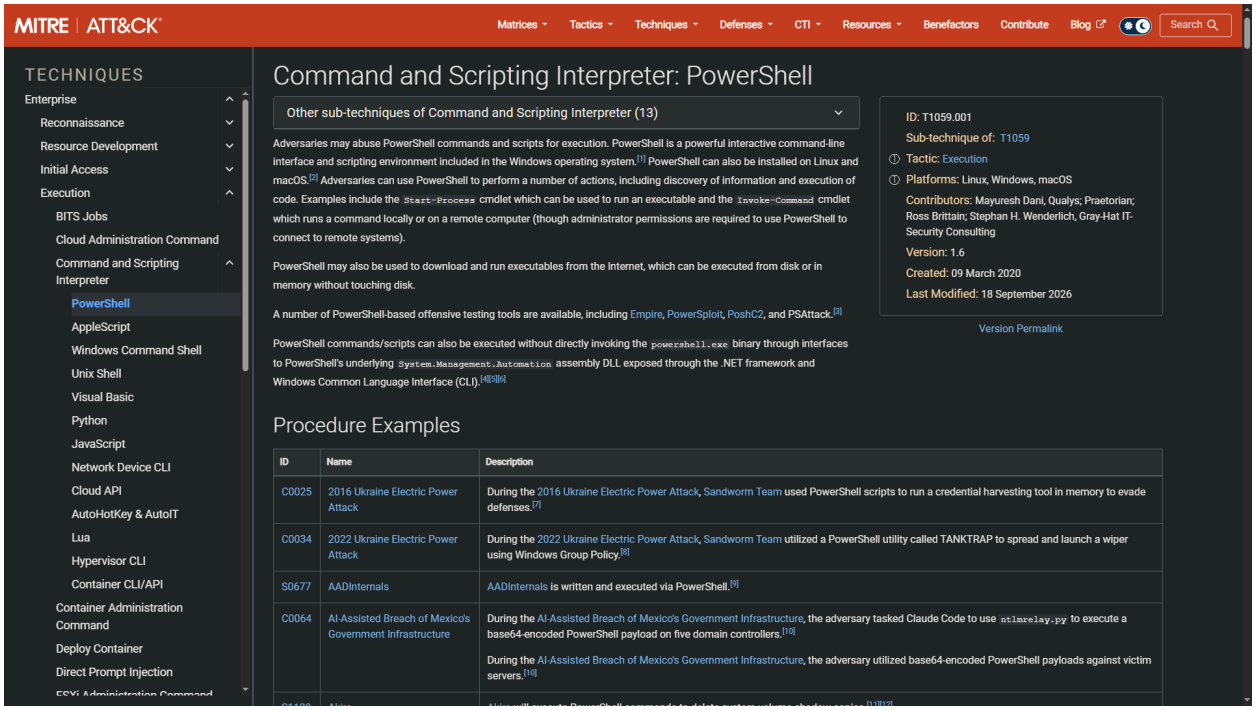
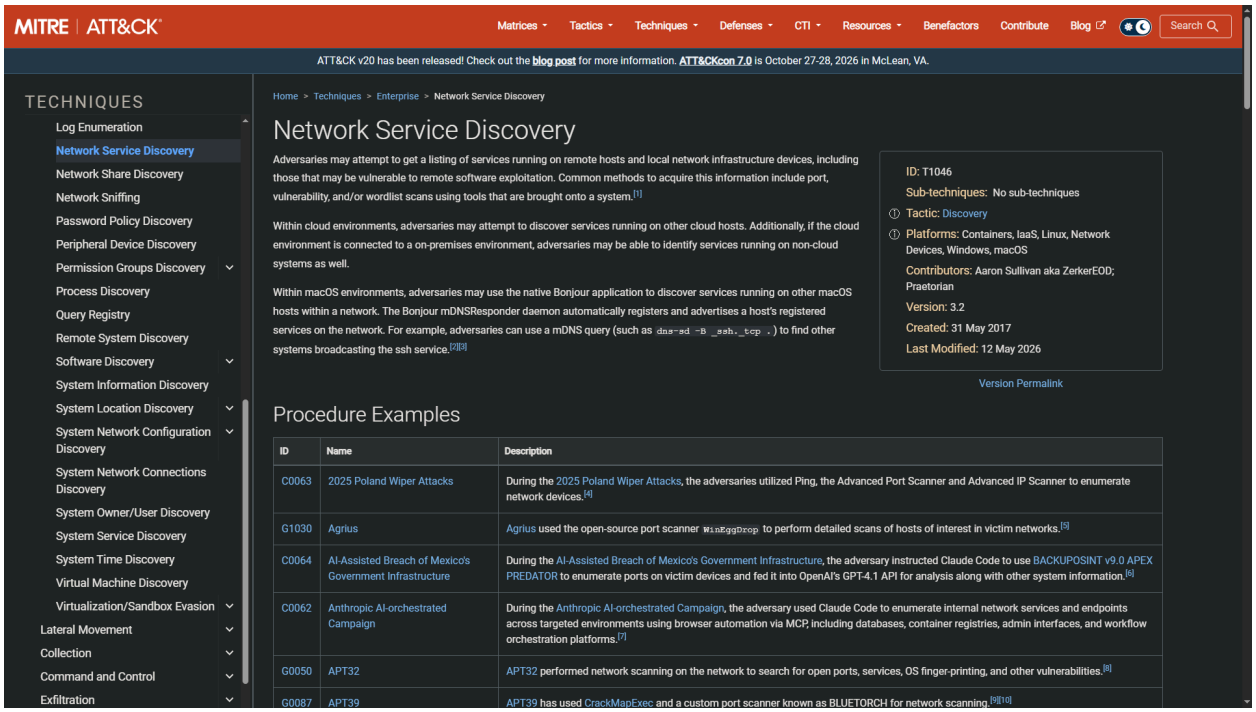

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TECHNIQUES
Hardware Additions
Phishing
Spearphishing Attachment
Spearphishing Link
Spearphishing via Service
Spearphishing Voice
Replication Through Removable Media
Supply Chain Compromise
Trusted Relationship
Valid Accounts
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Phishing
Sub-techniques (4)
Adversaries may send phishing messages to gain access to victim systems. All forms of phishing are electronically delivered social engineering. Phishing can be targeted, known as spearphishing. In spearphishing, a specific individual, company, or industry will be targeted by the adversary. More generally, adversaries can conduct non-targeted phishing, such as in mass malware spam campaigns.
Adversaries may send victims emails containing malicious attachments or links, typically to execute malicious code on victim systems. Phishing may also be conducted via third-party services (like social media platforms) or delivered via calendar invites. Phishing may also involve social engineering techniques, such as posing as a trusted source, as well as evasive techniques such as removing or manipulating emails or metadata/headers from compromised accounts being abused to send messages (e.g., Email Hiding Rules).[1][2] Another way to accomplish this is by Email Spoofing[3] the identity of the sender, which can be used to fool both the human recipient as well as automated security tools,[4] or by including the intended target as a party to an existing email thread that includes malicious files or links (i.e., "thread hijacking").[5]
Victims may also receive phishing messages that instruct them to call a phone number where they are directed to visit a malicious URL, download malware,[6][7] or install adversary-accessible remote management tools onto their computer (i.e., User Execution).[8]
ID: T1566
Sub-techniques: T1566.001, T1566.002, T1566.003, T1566.004
Tactic: Initial Access
Platforms: Identity Provider, Linux, Office Suite, SaaS, Windows, macOS
Contributors: Liora Itkin; Liran Ravich, CardinalOps; Ohad Zaidenberg, @ohad_mz; Philip Winther; Scott Cook, Capital One
Version: 2.8
Created: 02 March 2020
Last Modified: 18 September 2026
Version Permalink
Procedure Examples
ID
Name
Description
G1049
AppleJeus
AppleJeus has used spearphishing emails to distribute malicious payloads.
G0001
Axiom
Axiom has used spear phishing to initially compromise victims.[10][11]
G0115
GOLD SOUTHFIELD
GOLD SOUTHFIELD has conducted malicious spam (malspam) campaigns to gain access to victim's machines.[12]
S0009
Hikit
Hikit has been spread through spear phishing.[11]


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TECHNIQUES
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Phishing
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Spearphishing Link
Spearphishing via Service
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Phishing: Spearphishing Link
Other sub-techniques of Phishing (4)
Adversaries may send spearphishing emails with a malicious link in an attempt to gain access to victim systems. Spearphishing with a link is a specific variant of spearphishing. It is different from other forms of spearphishing in that it employs the use of links to download malware contained in email, instead of attaching malicious files to the email itself, to avoid defenses that may inspect email attachments. Spearphishing may also involve social engineering techniques, such as posing as a trusted source.
All forms of spearphishing are electronically delivered social engineering targeted at a specific individual, company, or industry. In this case, the malicious emails contain links. Generally, the links will be accompanied by social engineering text and require the user to actively click or copy and paste a URL into a browser, leveraging User Execution. The visited website may compromise the web browser using an exploit, or the user will be prompted to download applications, documents, zip files, or even executables depending on the pretext for the email in the first place.
Adversaries may also include links that are intended to interact directly with an email reader, including embedded images intended to exploit the end system directly. Additionally, adversaries may use seemingly benign links that abuse special characters to mimic legitimate websites (known as an "IDN homograph attack").[1] URLs may also be obfuscated by taking advantage of quirks in the URL schema, such as the acceptance of integer- or hexadecimal-based hostname formats and the automatic discarding of text before an "@" symbol: for example, hxxp://google.com@1157586937.[2]
Adversaries may also utilize links to perform consent phishing/spearphishing campaigns to Steal Application Access Tokens that grant immediate access to the victim environment. For example, a user may be lured into granting adversaries permissions/access via a malicious OAuth 2.0 request URL that when accepted by the user provide permissions/access for malicious applications.[3][4] These stolen access tokens allow the adversary to perform various actions on behalf of the user via API calls.[4]
Similarly, malicious links may also target device-based authorization, such as OAuth 2.0 device authorization grant flow which is typically used to authenticate devices without UIs/browsers. Known as "device code phishing," an adversary may send a link that directs the victim to a malicious authorization page where the user is tricked into entering a code/credentials that produces a device token.[5][6][7]
ID: T1566.002
Sub-technique of: T1566
Tactic: Initial Access
Platforms: Identity Provider, Linux, Office Suite, SaaS, Windows, macOS
Contributors: Jeff Sakowicz, Microsoft Identity Developer Platform Services (IDPM Services); Kobi Haimovich, CardinalOps; Mark Wee; Menachem Goldstein; Philip Winther; Saisha Agrawal, Microsoft Threat Intelligent Center (MSTIC); Shailesh Tiwary (Indian Army)
Version: 2.8
Created: 02 March 2020
Last Modified: 11 September 2026
Version Permalink


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Unix Shell
Visual Basic
Python
JavaScript
Network Device CLI
Cloud API
AutoHotKey & AutoIT
Lua
Hypervisor CLI
Container CLI/API
Container Administration Command
Deploy Container
Direct Prompt Injection
Command and Scripting Interpreter: PowerShell
Other sub-techniques of Command and Scripting Interpreter (13)
Adversaries may abuse PowerShell commands and scripts for execution. PowerShell is a powerful interactive command-line interface and scripting environment included in the Windows operating system.[1] PowerShell can also be installed on Linux and macOS.[2] Adversaries can use PowerShell to perform a number of actions, including discovery of information and execution of code. Examples include the Start-Process cmdlet which can be used to run an executable and the Invoke-Command cmdlet which runs a command locally or on a remote computer (though administrator permissions are required to use PowerShell to connect to remote systems).
PowerShell may also be used to download and run executables from the Internet, which can be executed from disk or in memory without touching disk.
A number of PowerShell-based offensive testing tools are available, including Empire, PowerSploit, PoshC2, and PSAttack.[3]
PowerShell commands/scripts can also be executed without directly invoking the powershell.exe binary through interfaces to PowerShell's underlying System.Management.Automation assembly DLL exposed through the .NET framework and Windows Common Language Interface (CLI).[4][5][6]
ID: T1059.001
Sub-technique of: T1059
Tactic: Execution
Platforms: Linux, Windows, macOS
Contributors: Mayuresh Dani, Qualys; Praetorian; Ross Brittain; Stephan H. Wenderlich, Gray-Hat IT-Security Consulting
Version: 1.6
Created: 09 March 2020
Last Modified: 18 September 2026
Version Permalink
Procedure Examples
ID
Name
Description
C0025
2016 Ukraine Electric Power Attack
During the 2016 Ukraine Electric Power Attack, Sandworm Team used PowerShell scripts to run a credential harvesting tool in memory to evade defenses.[7]
C0034
2022 Ukraine Electric Power Attack
During the 2022 Ukraine Electric Power Attack, Sandworm Team utilized a PowerShell utility called TANKTRAP to spread and launch a wiper using Windows Group Policy.[8]
S0677
AADInternals
AADInternals is written and executed via PowerShell.[9]
C0064
AI-Assisted Breach of Mexico's Government Infrastructure
During the AI-Assisted Breach of Mexico's Government Infrastructure, the adversary tasked Claude Code to use ntlmrelay.py to execute a base64-encoded PowerShell payload on five domain controllers.[10]
During the AI-Assisted Breach of Mexico's Government Infrastructure, the adversary utilized base64-encoded PowerShell payloads against victim servers.[10]


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TECHNIQUES
Log Enumeration
Network Service Discovery
Network Share Discovery
Network Sniffing
Password Policy Discovery
Peripheral Device Discovery
Permission Groups Discovery
Process Discovery
Query Registry
Remote System Discovery
Software Discovery
System Information Discovery
System Location Discovery
System Network Configuration Discovery
System Network Connections Discovery
System Owner/User Discovery
System Service Discovery
System Time Discovery
Virtual Machine Discovery
Virtualization/Sandbox Evasion
Lateral Movement
Collection
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Network Service Discovery
Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation. Common methods to acquire this information include port, vulnerability, and/or wordlist scans using tools that are brought onto a system.[1]
Within cloud environments, adversaries may attempt to discover services running on other cloud hosts. Additionally, if the cloud environment is connected to a on-premises environment, adversaries may be able to identify services running on non-cloud systems as well.
Within macOS environments, adversaries may use the native Bonjour application to discover services running on other macOS hosts within a network. The Bonjour mDNSResponder daemon automatically registers and advertises a host's registered services on the network. For example, adversaries can use a mDNS query (such as dns-sd -B _ssh._tcp .) to find other systems broadcasting the ssh service.[2][3]
ID: T1046
Sub-techniques: No sub-techniques
Tactic: Discovery
Platforms: Containers, IaaS, Linux, Network Devices, Windows, macOS
Contributors: Aaron Sullivan aka ZerkerEOD; Praetorian
Version: 3.2
Created: 31 May 2017
Last Modified: 12 May 2026
Version Permalink
Procedure Examples
ID
Name
Description
C0063
2025 Poland Wiper Attacks
During the 2025 Poland Wiper Attacks, the adversaries utilized Ping, the Advanced Port Scanner and Advanced IP Scanner to enumerate network devices.[4]
G1030
Agrius
Agrius used the open-source port scanner WinEggDrop to perform detailed scans of hosts of interest in victim networks.[5]
C0064
AI-Assisted Breach of Mexico's Government Infrastructure
During the AI-Assisted Breach of Mexico's Government Infrastructure, the adversary instructed Claude Code to use BACKUPOSINT v9.0 APEX PREDATOR to enumerate ports on victim devices and fed it into OpenAI's GPT-4.1 API for analysis along with other system information.[6]
C0062
Anthropic AI-orchestrated Campaign
During the Anthropic AI-orchestrated Campaign, the adversary used Claude Code to enumerate internal network services and endpoints across targeted environments using browser automation via MCP, including databases, container registries, admin interfaces, and workflow orchestration platforms.[7]
G0050
APT32
APT32 performed network scanning on the network to search for open ports, services, OS finger-printing, and other vulnerabilities.[8]
G0087
APT39
APT39 has used CrackMapExec and a custom port scanner known as BLUETORCH for network scanning.[9][10]